

Integrating MLflow with DagsHub and Creating a REST API

1. Background

DagsHub provides built-in integrations with MLflow, Git, and DVC, allowing you to track experiments, save and load models, and store all your model artifacts, metrics, and parameters remotely in one place.

2. Create a DagsHub Repository

1. Go to <https://dagshub.com/>
2. Create a new repository (e.g., 'mlflow-demo').
3. Clone it locally:

```
git clone https://dagshub.com/<username>/mlflow-demo.git
cd mlflow-demo
```

3. Set Up MLflow Tracking on DagsHub

Install dependencies:

```
pip install mlflow dagshub
```

Initialize tracking URI and credentials in your Python script:

```
import dagshub
import mlflow
import os

dagshub.init(repo_owner='<username>', repo_name='mlflow-demo', mlflow=True)

# Or manually set MLflow tracking URI
mlflow.set_tracking_uri("https://dagshub.com/<username>/mlflow-demo.mlflow")

# If the repository is private
os.environ['MLFLOW_TRACKING_USERNAME'] = '<username>'
os.environ['MLFLOW_TRACKING_PASSWORD'] = '<your_personal_access_token>'
```

4. Train and Log Model Example

Example with scikit-learn regression model:

```
import mlflow
import mlflow.sklearn
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error
import pandas as pd

data = pd.DataFrame({
    'x': range(1, 11),
    'y': [2*i + 1 for i in range(1, 11)]
})
X_train, X_test, y_train, y_test = train_test_split(data[['x']], data['y'], test_size=0.2, random_state=42)

with mlflow.start_run():
    model = LinearRegression()
    model.fit(X_train, y_train)

    preds = model.predict(X_test)
    mse = mean_squared_error(y_test, preds)

    mlflow.log_param("test_size", 0.2)
    mlflow.log_metric("mse", mse)
```

```

mlflow.sklearn.log_model(model, "model")

print("Model logged to DagsHub MLflow!")

```

5. Create REST API to Serve Model

Install requirements:

```

pip install fastapi uvicorn mlflow[extras]

```

Create 'app.py':

```

from fastapi import FastAPI
from pydantic import BaseModel
import mlflow

model_uri = "models:/<model_name>/Production"
model = mlflow.sklearn.load_model(model_uri)

app = FastAPI(title="MLflow Model API")

class InputData(BaseModel):
    x: float

@app.post("/predict")
def predict(data: InputData):
    prediction = model.predict([[data.x]])
    return {"prediction": prediction[0]}

```

Run the API:

```

uvicorn app:app --reload

```

Visit <http://127.0.0.1:8000/docs> to test interactively.

6. Example Requests

Using cURL:

```

curl -X POST "http://127.0.0.1:8000/predict" -H "Content-Type: application/json" -d '{"x": 5}'

```

Using Python:

```

import requests
response = requests.post("http://127.0.0.1:8000/predict", json={"x": 5})
print(response.json())

```

7. Summary

Step	Description
1	Create DagsHub repo
2	Initialize MLflow with DagsHub
3	Log model and metrics
4	Serve model using FastAPI
5	Make predictions through REST API